



1

Bearing Knowledge

Presented at the bearing damage course, WE 204
Presented by xxx
2025-05-29

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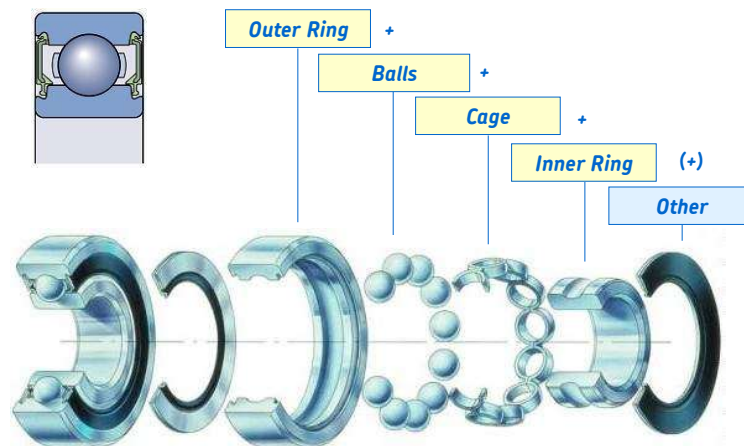
Bearing knowledge and terminology

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Bearing components – deep groove ball bearing



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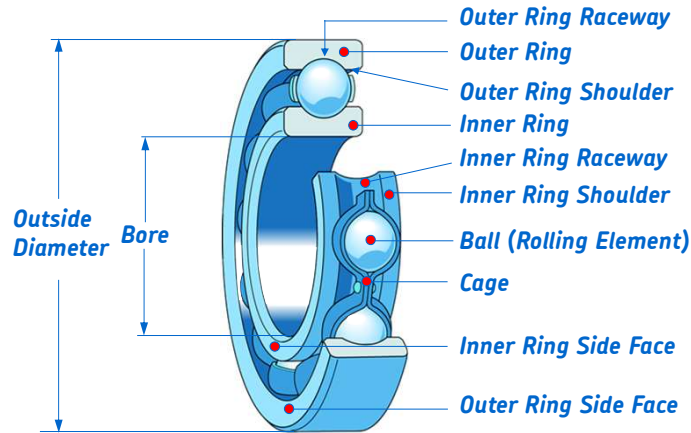
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Bearing terminology – deep groove ball bearing

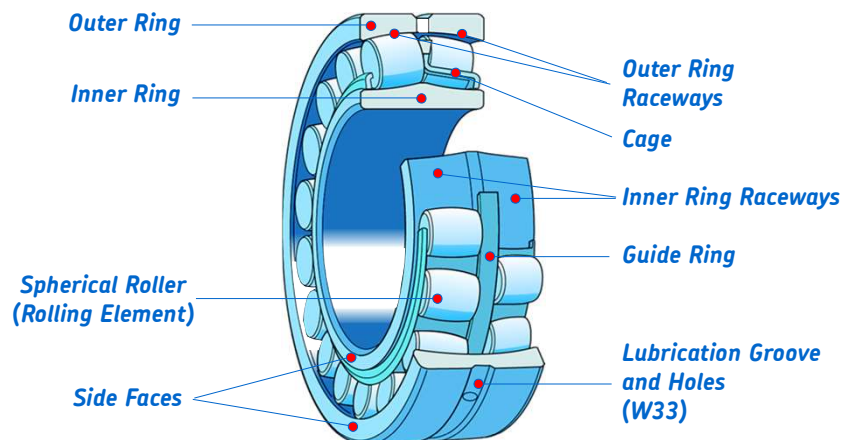


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Bearing terminology – spherical roller bearing



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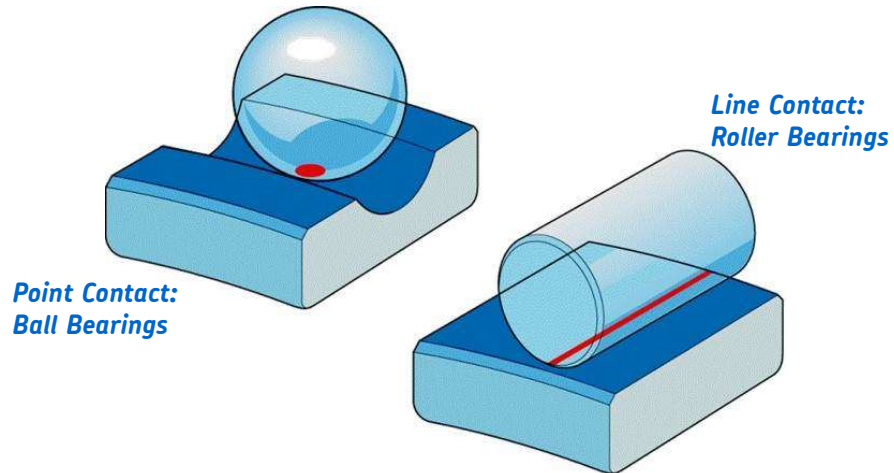
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Point and line contacts

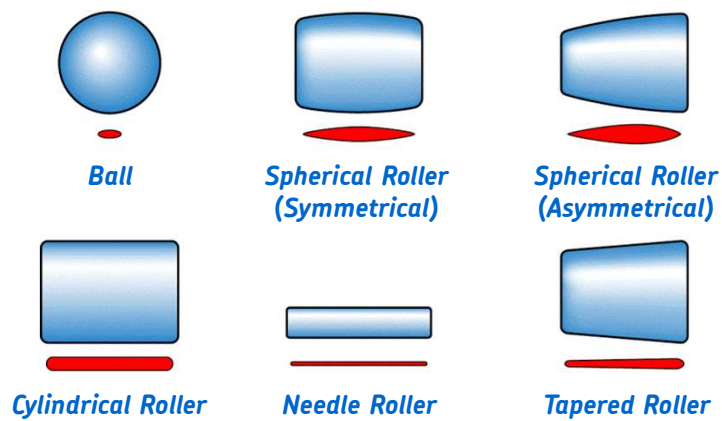


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Rolling element types



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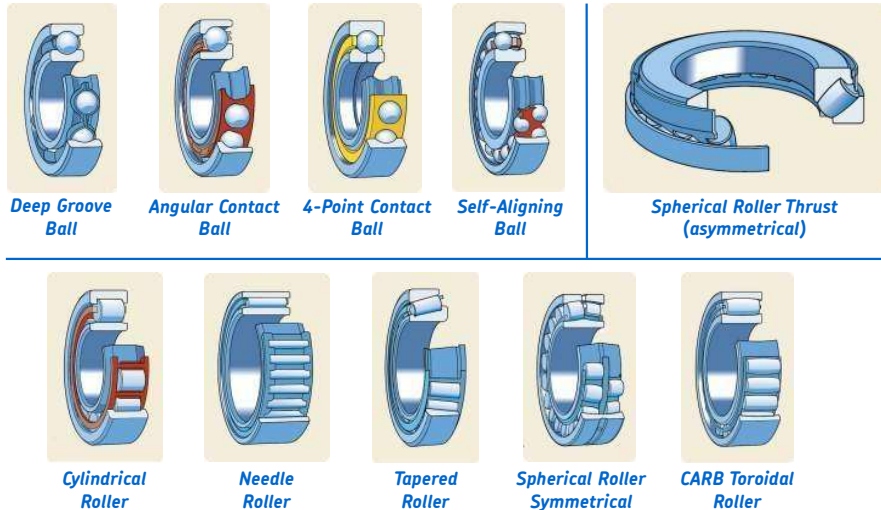
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Bearing types



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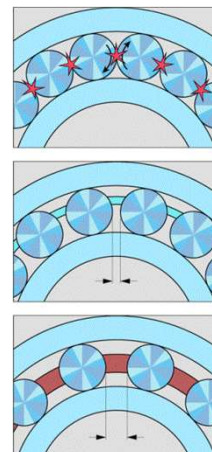
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The cage

Functions:

- Prevent contact between the rolling elements to minimize friction and heat generation
- Guide the rolling elements
- Provide space for lubricant
- Retain the rolling elements when bearings of separable design are mounted or dismounted



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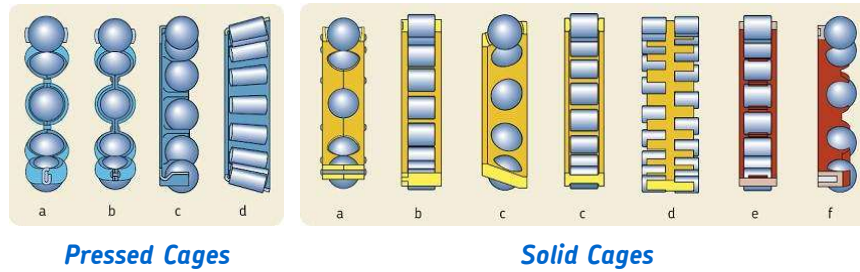
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Cage design and materials

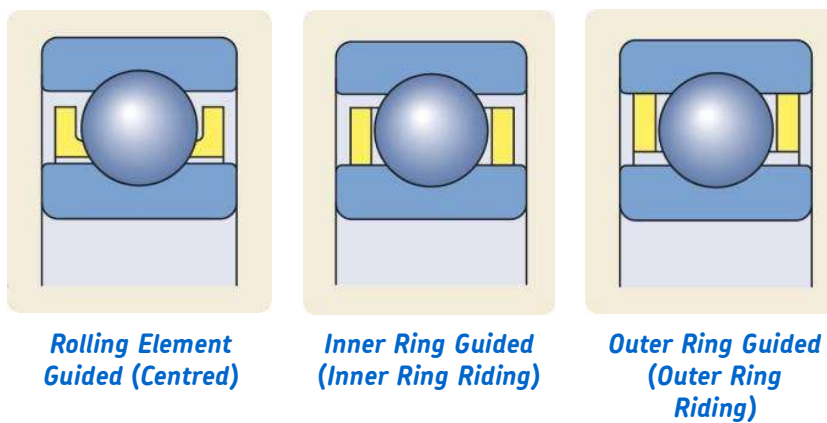


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Cage guidance (solid cages)



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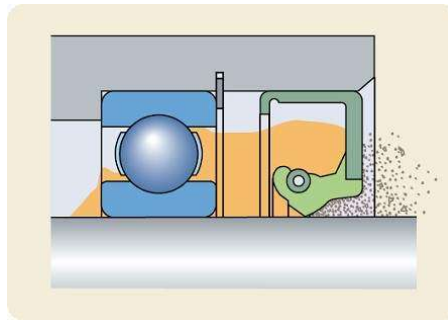
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Sealing arrangements

- Retain lubricant inside the bearing (arrangement)
- Keep contaminants out (the bearing arrangement)



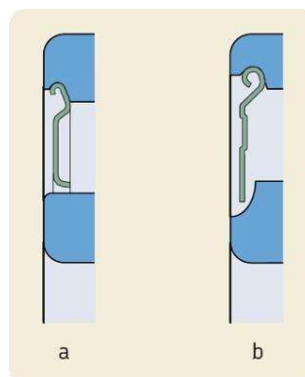
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Integral seals – shields (non-contact)

A long sealing gap with the land of the inner ring shoulder



An efficient labyrinth seal with a recess in the inner ring shoulder

Bearing with Shield
contamination is not heavy and where there is
no danger of water, steam, etc

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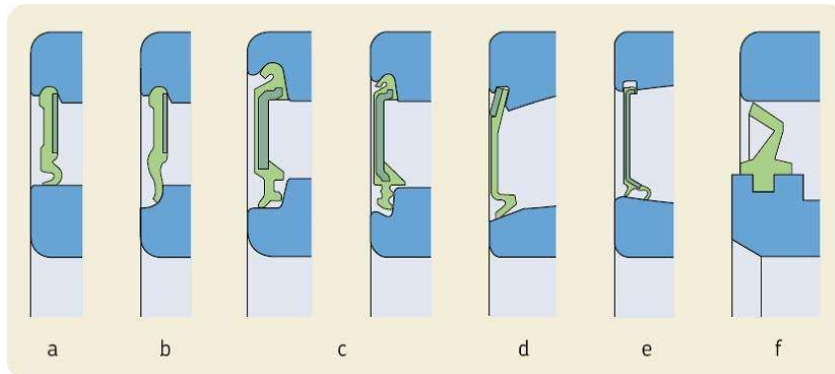
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Integral seals – seals (contact)



inner ring shoulder (a) and/or
against a recess in the inner ring
shoulder (b, c)

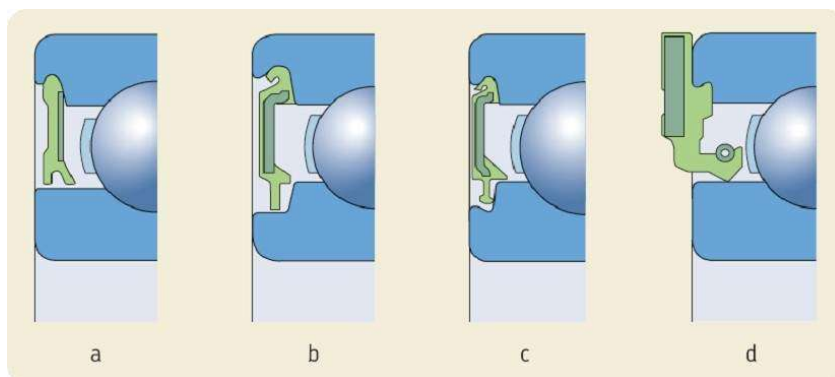
lead-in at the sides of the inner ring
raceway (d, e) or the outer ring (f)

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Integral seals – deep groove ball bearings



Non Contact
low-friction seal (a, b, c)

spring-loaded radial
shaft wavel seal (d)

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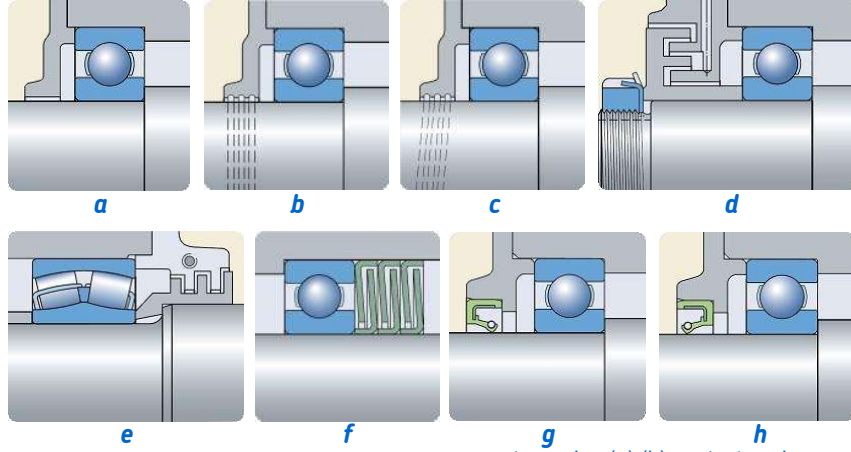
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External seals

Non contact gap-type seals (a) (b) (c)

multi-stage labyrinth seals (d) (e) (f)



garter spring (g) (h) contact seals
For oil lubricate bearing

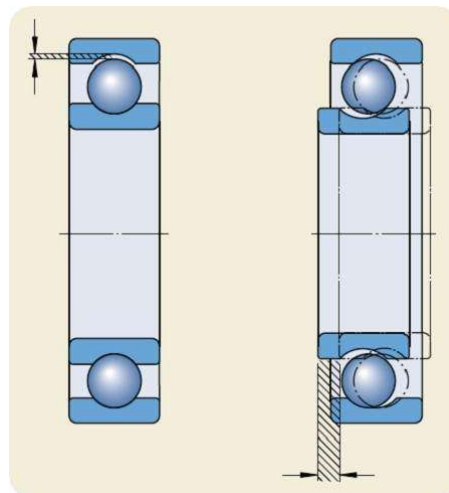
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Bearing internal clearance (radial and axial)

**Radial
Internal
Clearance**



**Axial
Internal
Clearance**

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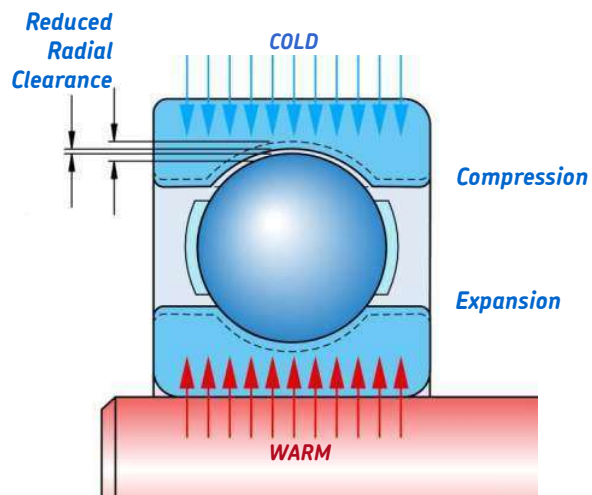
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Effect of temperature on radial clearance



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Bearing internal clearance

- **C1** Less than C2 clearance
- **C2** Less than Normal clearance
- **(CN)** Normal clearance (normally NOT marked)
- **C3** Greater than Normal clearance
- **C4** Greater than C3 clearance
- **C5** Greater than C4 clearance

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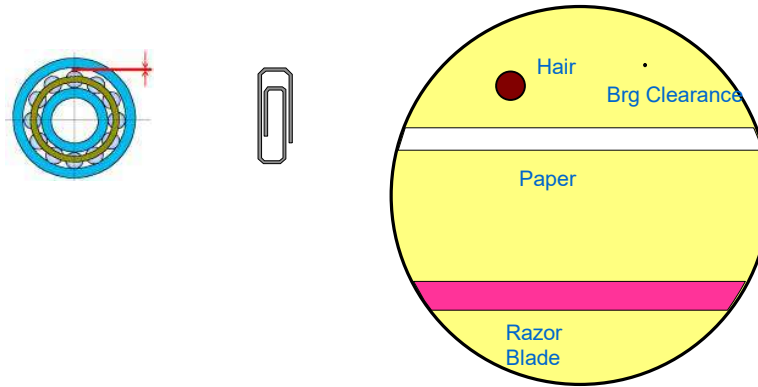
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Bearing operating clearance



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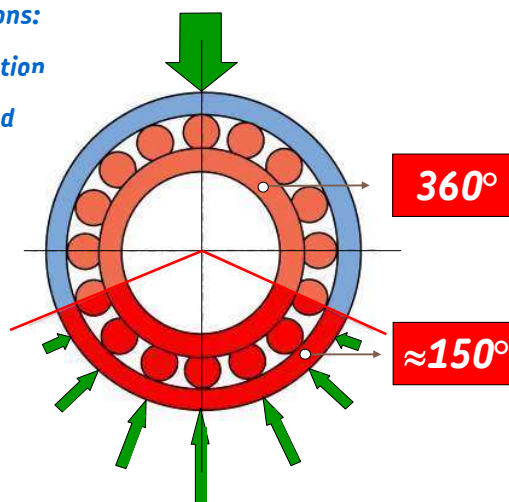
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Bearing load zone (inner ring rotation)

Operating conditions:

- Inner ring rotation
- Pure radial load



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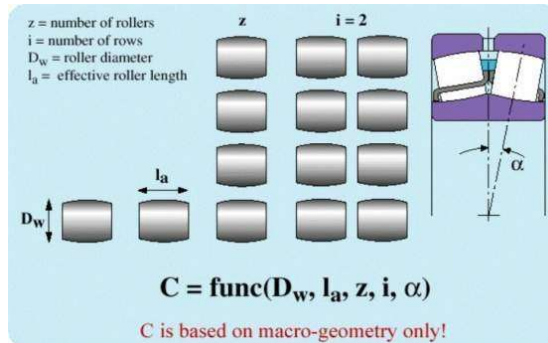
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Basic dynamic load rating

The basic dynamic load rating C is used for calculations involving dynamically stressed bearings, i.e., a bearing that rotates under load. It expresses the bearing load that will give an ISO 281:2007 basic rating life of 1,000,000 revolutions. It is assumed that the load is constant in magnitude and direction and is radial for radial bearings and axial, centrically acting, for thrust bearings.



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Bearing basic rating life (L_{10})

$$L_{10} = (C/P)^p$$

L_{10} : Basic rating life, 90% reliability (millions of revolutions)

C : Basic dynamic load rating, kN

P : Equivalent dynamic bearing load, kN

$()^p$: Exponent of the life equation

Theoretical Life :-

The life of a rolling bearing is defined as:

- The number of revolutions, or
- The number of operating hours at a given speed, which the bearing is capable of enduring before the first sign of metal fatigue (flaking, spalling) occurs on one of its rings or rolling elements.

Service Life :-

Represents the actual life of a bearing in real operating conditions before it fails.

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Specification life – guideline values

<i>Household machines, agriculture machines, and instruments</i>	300 to 3,000
<i>Intermittent use, hand tools, and construction machines</i>	3,000 to 8,000
<i>Elevators, cranes, and machines run for short periods with high reliability</i>	8,000 to 12,000
<i>Machines run 8 hours per day, electric motors for industrial use, general purpose gear boxes</i>	10,000 to 25,000
<i>Machine tools, woodworking machines, conveyor belts, printing equipment</i>	20,000 to 30,000
<i>Machines for use 24 hours per day: pumps, rolling mill gear drives, medium sized electric machinery</i>	40,000 to 50,000
<i>Ocean-going vessels, rotary furnaces, propulsion machinery</i>	60,000 to 100,000
<i>Large electric machinery, mine pumps and ventilator fans, power station equipment</i>	>100,000



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SKF rating life (L_{10m})

ISO 281:2007

$$L_{nm} = a_1 a_{SKF} (C/P)^p$$

L_{nm} : SKF rating life, (100-n) % reliability (millions of revolutions)

a_1 : life adjustment factor for reliability (1 for 90% reliability)

a_{SKF} : Life adjustment factor (based on SKF life method)
(in accordance with ISO 281:2007)

Accounts for lubrication conditions and contamination

C : Basic dynamic load rating, kN

P : Equivalent dynamic bearing load, kN

$()^p$: Exponent of the life equation



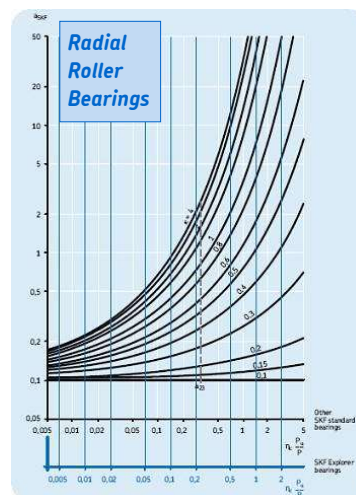
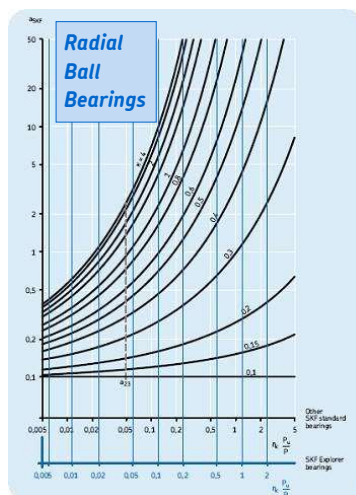
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$$a_{SKF} = f(\eta_c, P_u, P, \kappa)$$

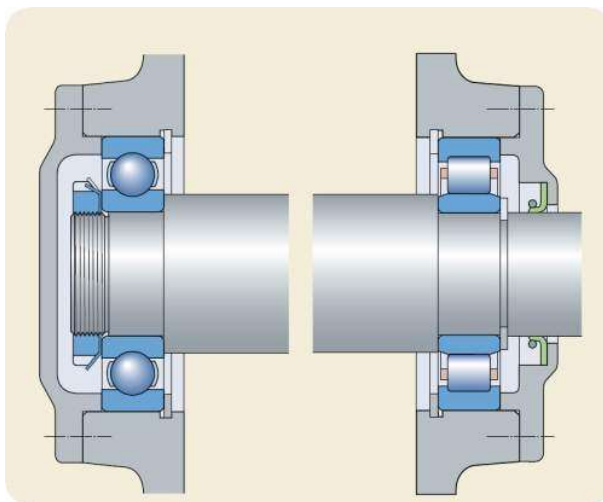


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Bearing systems – locating & non-locating bearings...



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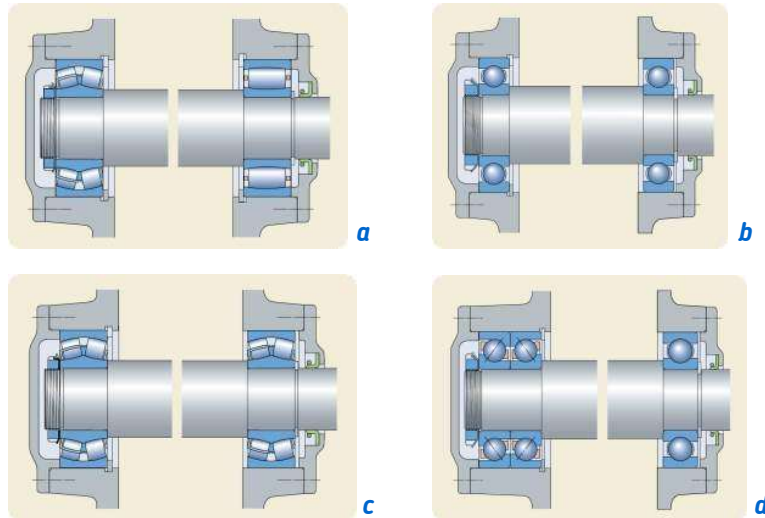
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Bearing systems – locating & non-locating bearings



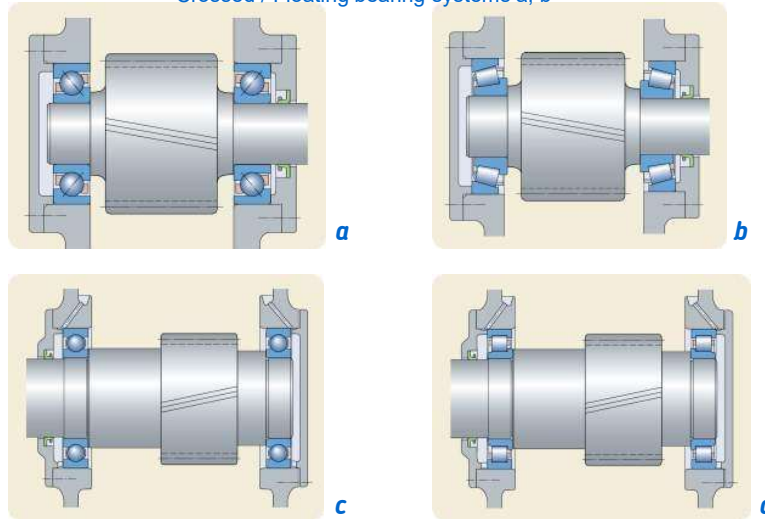
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Crossed and floating bearing systems

Crossed / Floating bearing systems a, b



"Floating" bearing systems c, d

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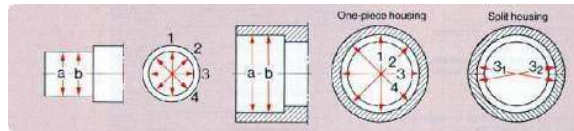
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Tolerances and fits – radial location of bearings



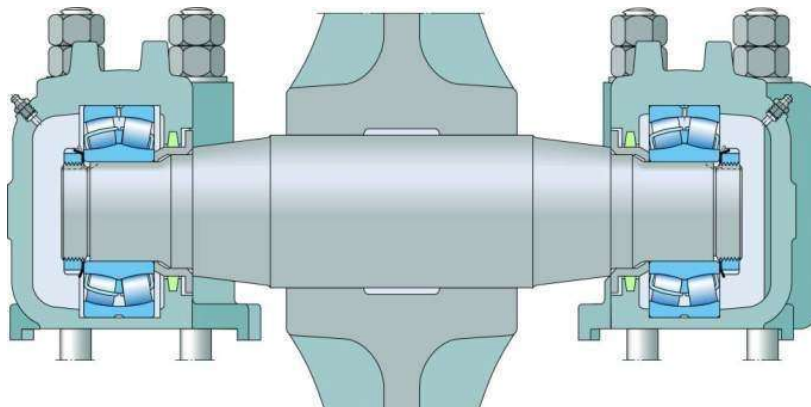
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Selection of fit – factors to consider...

1a. Condition of rotation – rotating load



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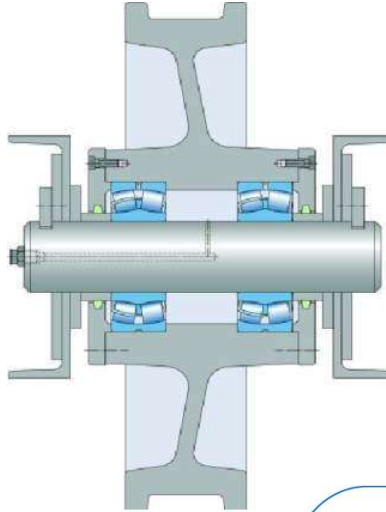
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Selection of fit – factors to consider...

1b. Condition of rotation – stationary load



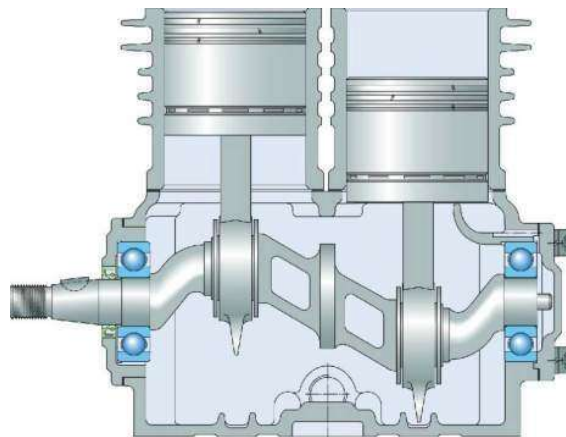
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Selection of fit – factors to consider...

1c. Condition of rotation – direction of load indeterminate



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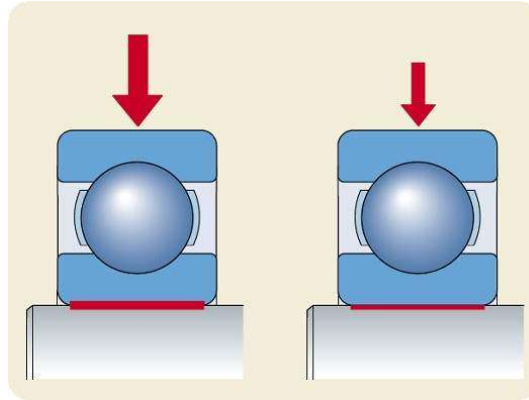
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Selection of fit – factors to consider...

2. Magnitude of load



Light Load: ($C/P=15$)

- $P \leq 0.05 C$

Normal Load: ($C/P=8$)

- $0.05 C < P \leq 0.1 C$

Heavy Load: ($C/P=4$)

- $0.1 C < P \leq 0.15 C$

Very Heavy Load: ($C/P=2$)

- $P > 0.15 C$

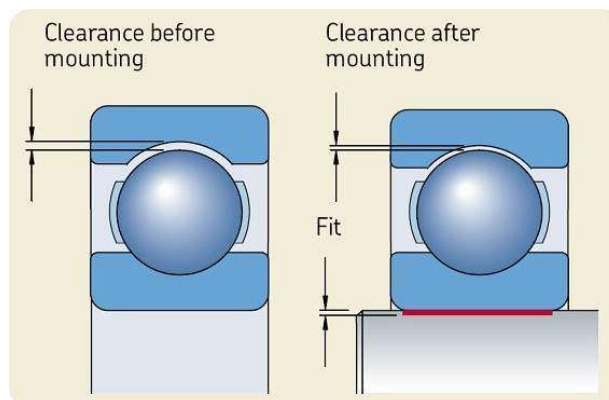
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Selection of fit – factors to consider...

3. Bearing Type and Bore Diameter



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Recommended fits – tables

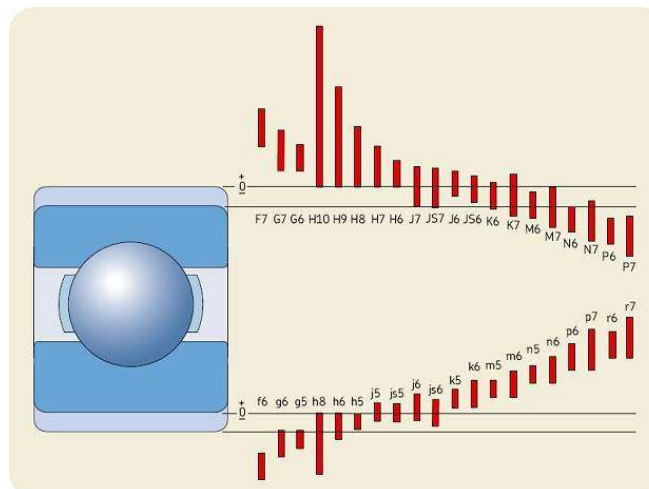
Fits for solid steel shafts					
Radial bearings with cylindrical bore					
Conditions ¹⁾	Examples	Shaft diameter, mm Ball bearings ¹⁾	Cylindrical roller bearings	Taper roller bearings	CARB and spherical roller bearings
Rotating inner ring load or direction of load indeterminate					
Light and variable loads ($P \leq 0,05 C$)	Conveyors, lightly loaded gearbox bearings	≤ 17 (17) to 100 (100) to 140	– ≤ 25 (25) to 60 (60) to 140	– ≤ 25 (25) to 60 (60) to 140	– – – –
Normal to heavy loads ($P > 0,05 C$)	Bearing applications generally, electric motors, turbines, pumps, gearing, wood-working machines, wind mills	≤ 10 (10) to 17 (17) to 100 – (100) to 140 (140) to 200 – (200) to 500 – > 500 – –	– – – ≤ 30 (30) to 50 – – (50) to 65 (65) to 100 (100) to 280 – (280) to 500 – > 500	– – – ≤ 40 – – – (40) to 65 (65) to 200 (200) to 360 – (360) to 500 – > 500	– – – – 25 to 40 – – (40) to 60 (60) to 100 (100) to 200 – (200) to 500 – > 500
Heavy to very heavy loads and shock loads with difficult working conditions ($P > 0,1 C$)	Axleboxes for heavy railway vehicles, traction motors, rolling mills	– – – – – –	– (50) to 65 (65) to 85 (85) to 140 (140) to 300 (300) to 500 – > 500	– (50) to 110 (110) to 200 (200) to 500 – – – > 500	– (50) to 70 – (70) to 140 (140) to 280 (280) to 400 – > 400
					$js5 (h5)^{2)}$ $js6 (h6)^{2)}$ $k6$ $m6$ $js5$ $js6 (h6)^{2)}$ $k5^{3)}$ $m5$ $m6$ $n5^{4)}$ $n6^{4)}$ $p6^{5)}$ $p7^{4)}$ $r6^{4)}$ $r7^{4)}$ $s6_{min} \pm (T6/2^{6)8)}$ $s7_{min} \pm (T7/2^{6)8)}$

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Recommended fits



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Effects of fits

Too loose:

- *Allows relative movement between mating components (bearing rotates on its seat)*
- *Results in: wear, heat, smearing, fretting, ring fracture...*

Too tight:

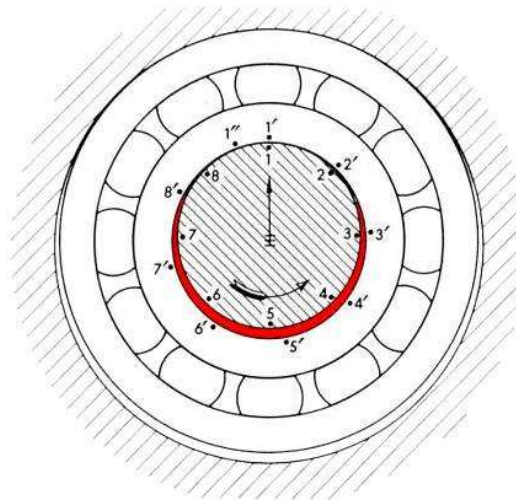
- *Too little clearance*
- *Increased operating temperature*
- *Too low operating viscosity*
- *Risk of ring cracking or fracture*

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Recommended fits: creep – effects of too loose a fit...



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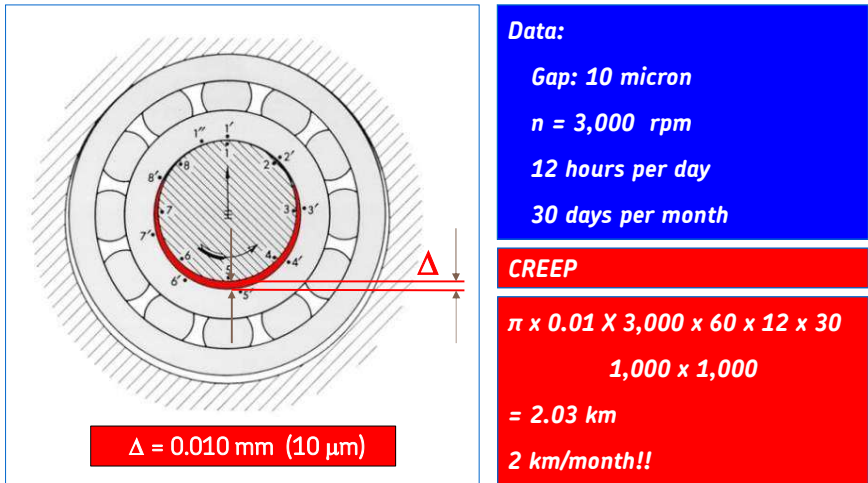
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Recommended fits: creep – effects of too loose a fit...



A creep of 2 km/month is quite a disaster. In the case of a bearing, it means wear, heating up, more wear, and possibility of failure.



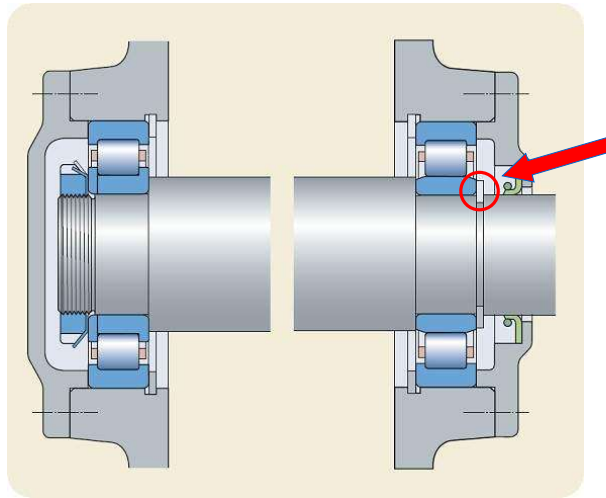
Recommended fits: creep – effects of too loose a fit



Result of very advanced creep



Axial location of bearings

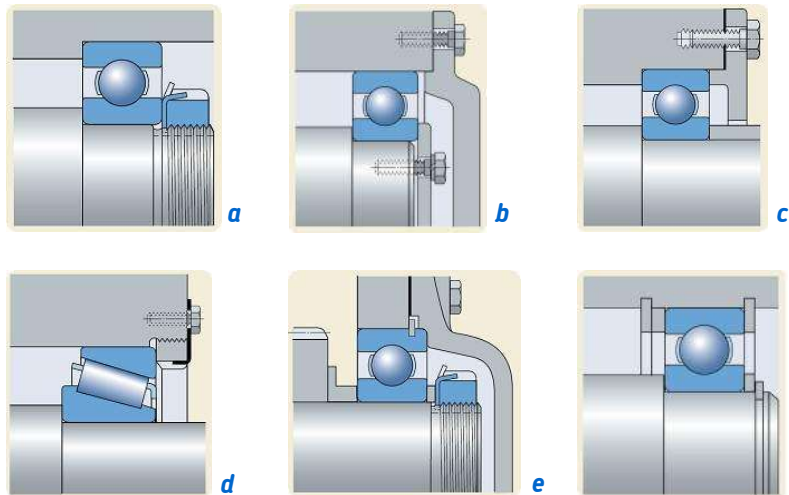


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Axial location – bearings with a cylindrical bore



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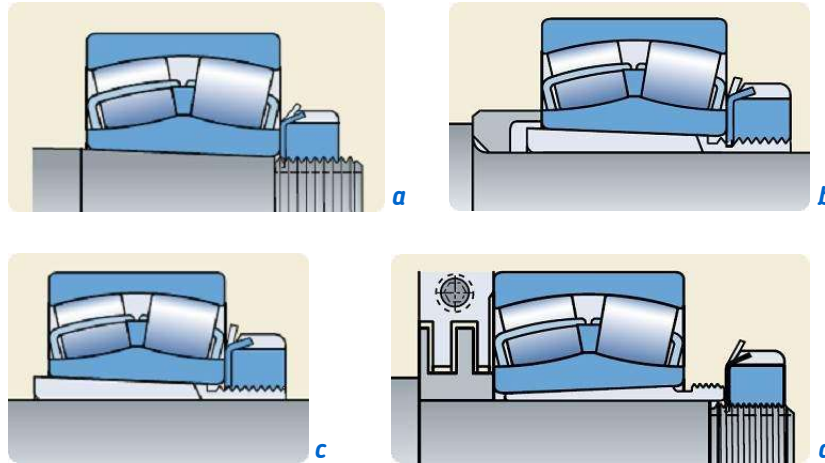
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Axial location – bearings with a tapered bore



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Thank you
Pre Test 30 Min

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